

Hybrid Control of Dual Active Bridge (DAB) Converter in EV Battery Charging for Performance Enhancement

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Abstract—In this paper, we have proposed a hybrid control technology for a DAB (Dual Active Bridge) converter for charging electric vehicles. It is designed to automatically shift from EPS (Extended Phase Shift) in CC(Constant Current) mode to DPS (Dual Phase Shift) in CV (Constant Voltage) mode. This method will improve charging performance by reducing switching loss and decreasing current stress, while extending the overall ZVS (Zero Voltage Switching) range.

Index Terms—DC to DC converter, DAB, Phase shift modulation, ZVS, SPS, DPS, EPS, CC–CV charging

I. INTRODUCTION

Among the different isolated bidirectional DC-DC converter configurations reported in literature, the DAB converter has gained considerable attention and practical adoption in electric vehicle charging infrastructure. It is also extensively employed in grid-connected renewable energy systems and energy storage platforms [1], [2]. The DAB topology is structured around a pair of active H-bridge networks, interconnected through a high-frequency isolation transformer, enabling safe and efficient power exchange between galvanically separated ports. This configuration enables bidirectional energy transfer with high power density and sustained efficiency across varying voltage and load conditions [3]–[5].

Rapid EV market expansion has intensified the need for chargers capable of maintaining high efficiency across diverse load and voltage operating conditions. EV battery charging typically progresses through two sequential stages: an initial CC phase followed by a CV phase. CC mode holds the current at a fixed high value for quick charging, while CV mode stabilizes the voltage as the battery reaches its limit. This wide range of operating conditions poses significant challenges for the converter, as a single control strategy may not deliver optimal performance across both charging modes [6]. In addition to control challenges, wide current and voltage excursions during CC–CV charging also increase stress on power devices and protection circuits, making robust and noise-immune overcurrent protection essential in SiC-based DAB converters [7].

Phase-shift modulation is commonly employed for controlling DAB converters due to its simplicity and effectiveness in maintaining Zero Voltage Switching (ZVS)

[8], [9]. SPS works well near unity voltage ratio, but struggles under light-load conditions where rising circulating current and limited power controllability cause efficiency to fall [10]. To address these drawbacks, researchers have proposed advanced modulation techniques such as DPS and EPS [11], [12]. In DPS modulation, internal phase shift of equal magnitude is introduced in both primary and secondary bridges. This shift extends Zero Voltage Switching (ZVS) range and enables independent control of active and reactive power. [10]. EPS employs asymmetric phase shifting, which improves soft-switching performance and reduces reactive current over a wide load range.

Despite the individual advantages of DPS and EPS, most existing studies apply a single modulation strategy throughout the entire charging process, limiting adaptability to varying operating conditions. EPS is more effective during the CC mode, where it reduces inductor current stress and reactive power, while DPS performs better during the CV mode by minimizing switching losses and maintaining ZVS under light-load conditions [13], [14]. Consequently, reliance on a single modulation method results in suboptimal efficiency, narrower ZVS coverage, and increased stress on power devices.

To address these issues, we have proposed an adaptive modulation framework, where the converter automatically selects the optimal phase-shift strategy based on the active charging stage. The proposed approach reduces inductor current stress, lowers switching losses, and broadens the ZVS operating range, ensuring smooth transitions between charging stages and improved overall efficiency. Simulation results obtained using PLECS validate that the proposed hybrid control scheme outperforms conventional single-mode strategies. These results confirm its viability for high-frequency isolated onboard EV charger designs.

II. DAB CONVERTER TOPOLOGY

A. Equivalent Circuit and Power Transfer Principle

A basic DAB converter is presented in Fig. 1, where the primary and secondary sides are kept electrically isolated through a high-frequency transformer. Here, L denotes the transformer's leakage inductance, which may also be realized

as an external series inductor if needed, acting as the primary energy transfer element. This inductance also contributes to soft switching and governs the power-transfer capability, inductor dynamics of the converter [14].

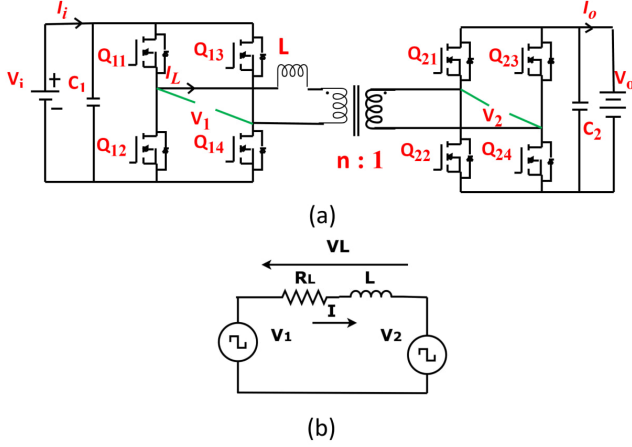


Fig. 1. DAB converter: (a) full circuit topology, (b) simplified equivalent circuit.

Fig. 1(b) presents the simplified equivalent circuit, where both H-bridges of the DAB are modeled as square wave voltage sources. Here, V_1 denotes the primary side voltage, while V_2 is the secondary side voltage reflected to the primary. The leakage inductance voltage V_L is then given by:

$$V_L = V_1 - V_2, \quad (1)$$

This voltage drives the inductor current waveform and governs both the instantaneous and mean power transferred. Output power is controlled by varying the phase shift between primary and secondary bridge voltages, which facilitates both bidirectional energy flow and soft-switching behavior [11]. All H-bridges operate with a 50% duty-cycle complementary gating scheme to prevent shoot-through. Inductor current waveform is shaped by the voltage across the inductance, which in turn dictates both the instantaneous and cycle-averaged power delivery. By tuning the phase angle between the two bridge output voltages, both bidirectional energy transfer and soft-switching can be simultaneously achieved [11]. Nearly all DAB converters operate near a 50% duty cycle to prevent cross-conduction. Two types of phase shifts are employed:

- **Inter-bridge phase shift ϕ :** Phase offset between the primary and secondary bridge voltage fundamentals.
- **Intra-bridge phase shift δ (δ_1, δ_2):** Leg-to-leg phase difference within the same H-bridge (denoted as δ_1, δ_2), used to increase the degree of power control.

Unless otherwise stated, all phase-shift angles are expressed in **radians**.

1) *Single Phase Shift (SPS)*: In SPS, the amount of power transferred is controlled by adjusting the inter-bridge phase

shift. The resulting inductor current, determined by inductor voltage, has a triangular waveform with defined peak values.

$$I_1 = \frac{T_s}{4L} \left[V_i + nV_o \left(\frac{2\phi}{\pi} - 1 \right) \right], \quad (2)$$

$$I_2 = \frac{T_s}{4L} \left[nV_o - V_i \left(\frac{2\phi}{\pi} - 1 \right) \right]. \quad (3)$$

The average output power and current are given by

$$P_o^{\text{SPS}} = \frac{nV_i V_o}{2\pi^2 L f_s} (\pi - \phi) \phi, \quad (4)$$

$$I_o = \frac{nV_i}{2\pi^2 L f_s} (\pi - \phi) \phi. \quad (5)$$

Peak power occurs at $\phi = \pi/2$. Despite its simplicity and effectiveness in achieving ZVS under unity voltage conversion ratio, SPS modulation is susceptible to voltage mismatch, which amplifies circulating current, intensifies device stress, and limits the achievable soft-switching coverage. Fig. 3 demonstrates that such performance degradation intensifies under light-load operation, most notably near the conclusion of the CC charging stage. These limitations have motivated the development of more advanced strategies, such as EPS and DPS modulation, which are discussed below.

2) *Extended Phase Shift (EPS)*: In EPS, control signal of a specific leg on primary side of DAB is shifted to create an intra-bridge phase shift, while secondary-side side remains unshifted. This additional degree of freedom creates partial freewheeling intervals that reduce circulating current compared to SPS. As shown in Fig. 2, the modified primary switching sequence generates flat-topped segments in the inductor current profile, suppressing RMS current stress relative to SPS. Resulting current slopes I_1 , I_p , and I_2 are expressed as

$$I_1 = \frac{T_s}{4L} \left[V_i \left(1 - \frac{\delta}{\pi} \right) + nV_o \left(\frac{2\delta}{\pi} + \frac{2\phi}{\pi} - 1 \right) \right], \quad (6)$$

$$I_p = \frac{T_s}{4L} \left[V_i \left(1 - \frac{\delta}{\pi} \right) + nV_o \left(\frac{2\phi}{\pi} - 1 \right) \right], \quad (7)$$

$$I_2 = \frac{T_s}{4L} \left[V_i \left(\frac{2\phi}{\pi} + \frac{\delta}{\pi} - 1 \right) + nV_o \right]. \quad (8)$$

The normalized phase-shift constraints are defined as

$$0 \leq \phi \leq \pi, \quad 0 \leq \delta \leq \pi, \quad 0 \leq \phi + \delta \leq \pi.$$

The average transmitted power under EPS modulation is given by

$$P_o = \frac{nV_i V_o T_s}{2L} \left[\left(1 - \frac{\phi}{\pi} \right) \frac{\phi}{\pi} + \frac{1}{2} \frac{\delta}{\pi} \left(1 - \frac{\delta}{\pi} - \frac{2\phi}{\pi} \right) \right]. \quad (9)$$

EPS significantly enhances efficiency during the CC charging stage, where high current and robust ZVS are required by curtailing circulating current and enhancing soft switching range [15].

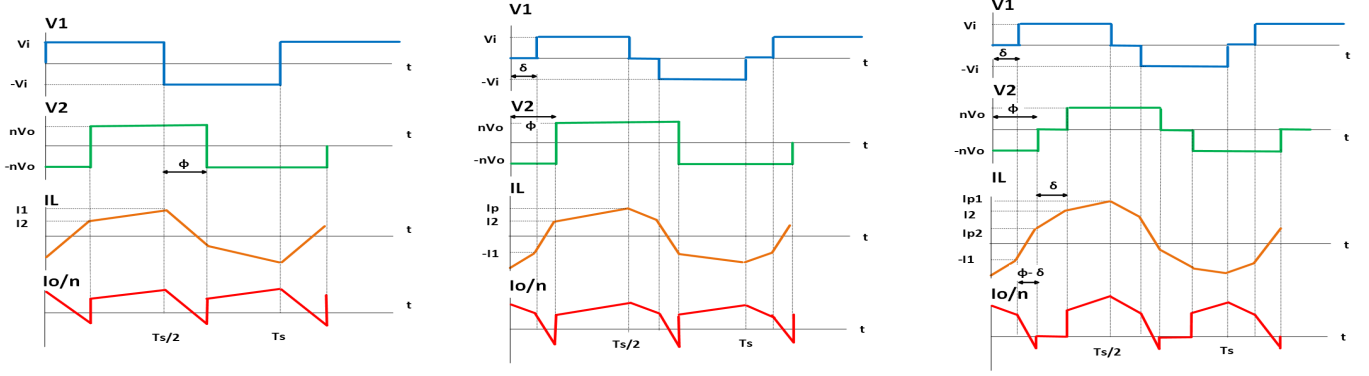


Fig. 2. Primary and secondary voltage and current waveforms under (a) SPS, (b) EPS, and (c) DPS

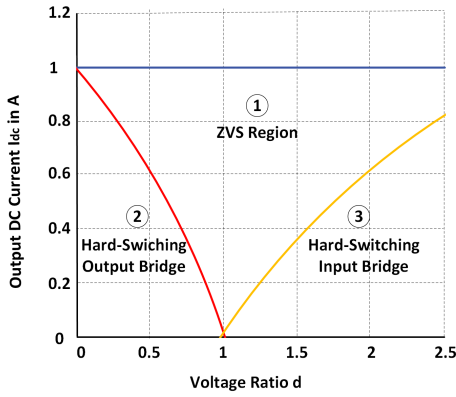


Fig. 3. ZVS operation of the DAB converter under different voltage ratios and load currents

3) *Dual Phase Shift (DPS)*: By symmetrically extending the internal phase shift δ to both converter bridges, DPS builds upon the EPS strategy to achieve a more balanced control over power transfer. It introduces two independent control variables, ϕ and δ , enabling finer regulation of power transfer and circulating current. The resulting symmetric waveforms, shown in Fig. 2, generate freewheeling intervals on both sides of the transformer, producing a more balanced inductor current profile and further reducing reactive current compared to SPS and EPS.

When $|\phi| < \delta$, the inductor current values at commutation are given by

$$I_1 = I_{p2} = \frac{T_s}{4L} \left[V_i \left(1 - \frac{\delta}{\pi}\right) + nV_o \left(\frac{\delta}{\pi} - 1\right) \right], \quad (10)$$

$$I_2 = \frac{T_s}{4L} \left[V_i \left(\frac{\delta}{\pi} + \frac{2\phi}{\pi} - 1\right) + nV_o \left(1 - \frac{\delta}{\pi}\right) \right], \quad (11)$$

$$I_{p1} = \frac{T_s}{4L} \left[V_i \left(1 - \frac{\delta}{\pi}\right) + nV_o \left(\frac{\delta}{\pi} + \frac{2\phi}{\pi} - 1\right) \right]. \quad (12)$$

The symmetry of DPS modulation reduces RMS current and device stress, making it particularly suitable for the CV charging stage, where load current is low and minimizing

circulating power is essential for maintaining high efficiency [15].

B. Zero-Voltage Switching (ZVS)

ZVS is an important characteristic of the DAB that enables high efficiency by significantly reducing switching losses at high switching frequencies. Owing to the energy storage capability of the leakage inductor, MOSFET internal capacitances are charged and discharged during dead time, enabling zero voltage across the MOSFETs at turn-on. ZVS depends on instantaneous inductor current $I_L(t)$, phase-shift parameters ϕ and δ , and voltage ratio $k = \frac{V_o}{V_i}$. Under light-load conditions, transformer magnetizing current, together with inductor current, helps achieve ZVS; in most other cases, inductor current alone is sufficient to maintain ZVS [11], [16].

The fundamental ZVS condition can be expressed as

$$\frac{1}{2} L I_L^2 \geq \frac{1}{2} C_{eq} V^2, \quad (13)$$

where the inductor must store adequate energy to completely commutate the switch-node voltage through C_{eq} before the gate signal arrives C_{eq} .

Sufficient dead time is inserted in the gate signals before each switch turn-on to ensure that the inductor current can drive the switch-node voltage to zero. The required dead time can be estimated as [3]

$$t_{dead} = \frac{1}{f_s} \frac{nV_i + V_o(2\phi - 1)}{4(nV_i + V_o)}, \quad (14)$$

with f_s being the gate switching frequency, V_i and V_o denoting the DC-link voltages at the primary and secondary terminals, n capturing the voltage scaling ratio of the isolation transformer, and ϕ representing the inter-bridge phase angle governing power flow direction and magnitude.

III. PROPOSED CONTROL FRAMEWORK FOR CC–CV OPERATION

To achieve efficient CC–CV battery charging across the full operating range, the proposed framework selectively employs EPS and DPS modulation within a unified DAB

converter architecture.. The hybrid approach leverages EPS for high-efficiency CC operation and DPS for low-current CV operation, reducing circulating power, enhancing ZVS, and maintaining precise current and voltage control across the entire charging process.

In the proposed hybrid EPS–DPS control scheme:

- **EPS in CC mode** introduces asymmetric intra-bridge phase shifts, generating negative inductor current at appropriate intervals to enhance ZVS at high currents.
- **DPS in CV mode** uses balanced phase shifts to reduce circulating currents while preserving ZVS at low and moderate loads.

A. Battery Charging System Using DAB Converter

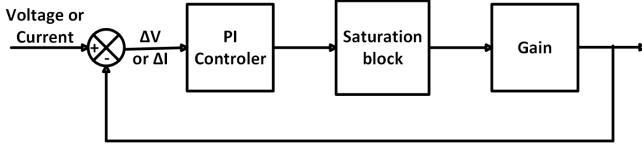


Fig. 4. PI Controller Design for Current and Voltage Regulation in CC- CV Battery Charging

Control architecture for both charging stages is shown in Fig. 4. CC controller regulates the battery current by adjusting the EPS phase-shift variables through a PI controller. The PI regulator is derived from a simplified small-signal model based on SPS, while its output is mapped to EPS parameters to reduce circulating current and maintain robust ZVS. This approach ensures high efficiency during the fast-charging mode when current demand is high.

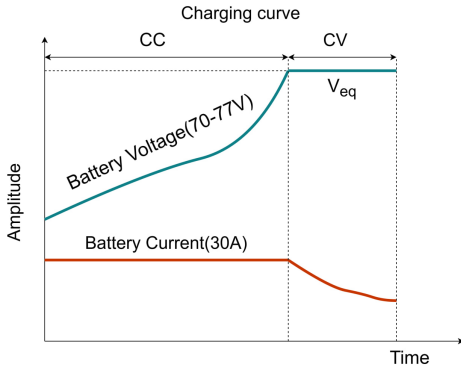


Fig. 5. Battery charging profile using the CCCV algorithm

Initially, the battery is charged in CC mode up to about 80% SoC. It then transitions from CC to CV, as shown in Fig. 5. Charging current is kept at a low value in CV mode, preventing overcharge and ensuring battery longevity [3], [13].

B. PI Controller

A single three-point PI controller structure is utilized to manage both CC and CV operation. In CC mode, the controller acts on the current error signal, whereas in CV mode it acts on

the voltage error signal. A supervisory controller determines the active control mode based on battery SOC and terminal voltage, enabling smooth transitions between CC and CV modes.

C. Loss Calculation

To quantify DAB efficiency and thermal performance, a loss analysis of MOSFET components is required [17]. Generally, two fundamental loss mechanisms are considered in MOSFETs .

1) *Conduction Losses*: Each MOSFET produces heat when current passes through its internal resistance during the on-state, causing conduction losses expressed as:

$$P_{\text{cond}} = I_{\text{Mos,rms}}^2 R_{\text{DS(on)}}, \quad (15)$$

where $I_{\text{Mos,rms}}$ is,

$$I_{\text{Mos,rms}} = \sqrt{\frac{1}{T_s} \int_0^{T_s} i_{\text{Mos}}^2(t) dt}. \quad (16)$$

and $R_{\text{DS(on)}}$ is the MOSFET on-resistance. This approach is widely used in high-frequency isolated converters to estimate conduction losses.

2) *Switching Losses*: Switching loss arises when non-zero voltage and current overlap during MOSFET turn-on and turn-off transitions. This loss is approximated as:

$$P_{\text{sw}} = \frac{1}{2} V_{\text{DC}} I_{\text{pk}} (t_{\text{on}} + t_{\text{off}}) f_s, \quad (17)$$

where V_{DC} is the DC-link voltage, I_{pk} is the peak switching current, t_{on} and t_{off} are the switching transition times, and f_s is the switching frequency. ZVS operation can significantly reduce turn-on losses, but switching loss becomes load-dependent under light-load conditions [18].

3) *Total Losses*: The overall power loss of the DAB converter is the sum of both conduction and switching losses across all MOSFETs:

$$P_{\text{total}} = P_{\text{cond}} + P_{\text{sw}}. \quad (18)$$

forming the basis for efficiency estimation and thermal design in MOSFET-only DAB converters for EV charging applications.

IV. SIMULATION RESULTS

Circuit parameters used in this study are summarized in Table I. PLECS-based time-domain simulations are carried out with a PI regulator governing the output quantity in each charging stage. The converter delivers a regulated current to the battery until a predefined terminal voltage threshold is reached, beyond which output voltage is held constant while current decays naturally. To achieve optimal performance in each stage, the phase-shift control variables are tuned independently, prioritizing loss reduction and reactive current minimization.

TABLE I
CIRCUIT PARAMETERS

Parameter	Value
Supply Voltage V_i	300 V
Load side Voltage V_2	80 V
Inductance L	15 μ H
Switching Frequency f_s	20 kHz
Transformer Ratio ($n_p:n_s$)	30:8
MOSFET Model	IMBG65R022M1H
R_{on}	22 m Ω

A. CC Mode Operation (EPS vs. SPS)

CC mode requires high output current, resulting in large inductor ripple and increased conduction stress. That's why EPS method is used in CC mode, as it reduces RMS current as well as conduction losses. In the simulation model, EPS operates with an external phase shift of $\phi = 11^\circ$ along with an internal phase shift of $\delta = 18^\circ$. This internal shift effectively suppresses circulating current while maintaining the required power transfer. As illustrated in Fig. 8, EPS consistently achieves higher efficiency across the CC operating range.

For an output power of 2193.7 W, the RMS inductor current and total converter losses show clear improvement:

- SPS: $i_{L,rms} = 13.14$ A, loss = 119 W
- EPS: $i_{L,rms} = 12.40$ A, loss = 103 W

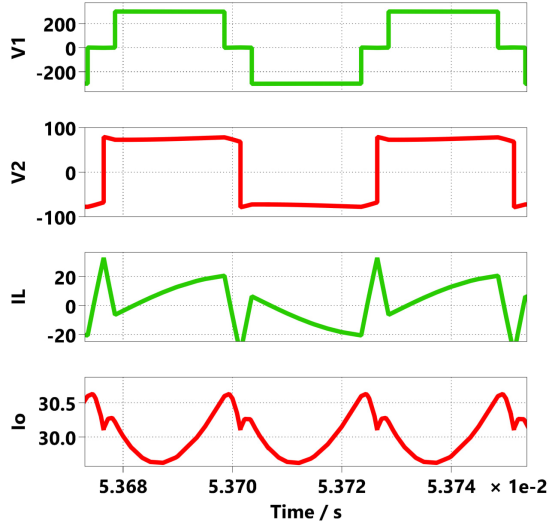


Fig. 6. Transformer voltage and current with output load current in CC mode (EPS).

The corresponding EPS waveform in Fig. 6 highlights the reduced current stress enabled by the internal phase shift, confirming EPS as the preferred modulation method for current-regulated CC operation.

B. CV Mode Operation (DPS/SPS)

In the CV charging stage, voltage is kept at a steady value as the battery current continuously drops toward zero. Under these light-load conditions, the inductor ripple naturally diminishes, leading SPS and DPS to exhibit nearly identical

RMS current. In the simulation, DPS operates with an external phase shift of $\phi = 2.57^\circ$ and the same internal phase shift $\delta = 18^\circ$ used in EPS. As shown in Fig. 9, EPS becomes less efficient in CV mode due to higher circulating current and a reduced soft-switching window. In contrast, SPS and DPS maintain higher efficiency across the entire CV region. At very low current, DPS achieves a slight advantage over SPS because the symmetric phase-shift structure further reduces residual reactive power, even when load current is small.

Overall, either SPS or DPS is suitable for voltage-regulated operation, with DPS offering marginally better efficiency, lower ripple, and a broader ZVS range. A combined strategy—EPS during CC mode and DPS during CV mode—therefore provides reduced total loss and improved charging performance.

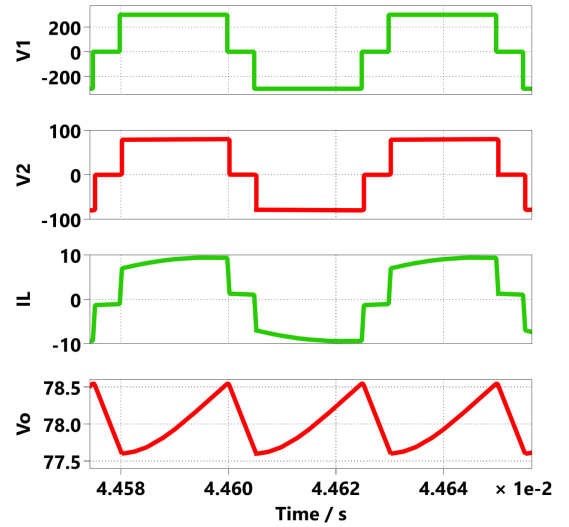


Fig. 7. Transformer voltage and current with output load current in CV mode (DPS).

C. Loss Calculation and Comparison

Loss estimation follows standard MOSFET conduction and switching-loss models [18]. Table II summarizes the analytical and simulated loss results for CC (EPS) and CV (DPS) modes.

TABLE II
ANALYTICAL VS. SIMULATED LOSSES COMPARISON

Mode	Analytical (W)	Simulated (W)	Error (%)
CC (EPS)	106.38	103.20	+3.08%
CV (DPS)	40.3	41.50	-8.06%

The results verify that EPS is preferred in CC mode due to lower circulating current, reduced RMS current, and minimized conduction losses. In CV mode, SPS and DPS experience inherently lower current stress; however, DPS retains a slight efficiency advantage at low current because of reduced residual reactive power. The analytical predictions show small deviations (3–8%) from simulation, consistent with common modeling approximations for switching.

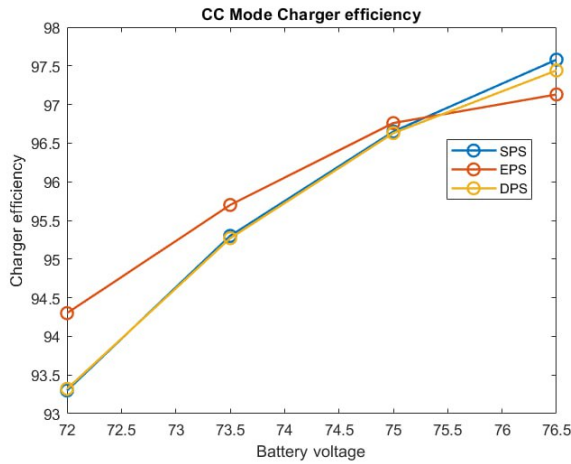


Fig. 8. CC mode charger efficiency comparison for SPS, EPS, and DPS.

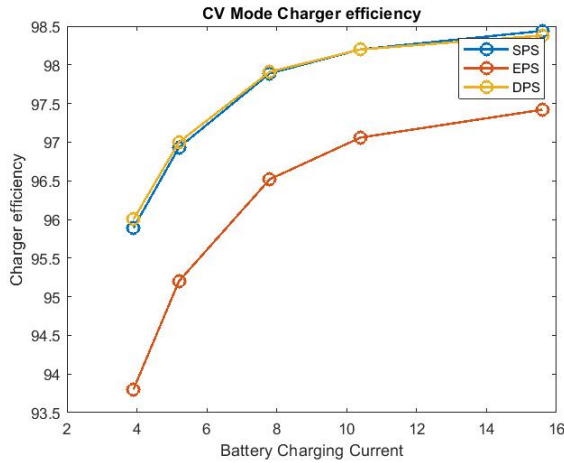


Fig. 9. CV mode charger efficiency comparison for SPS, EPS, and DPS.

V. CONCLUSION

This Paper examined a mode-adaptive phase-shift control scheme for a DAB-based EV battery charger, where EPS and DPS modulations are selectively engaged based on the active charging stage to achieve optimal converter performance. EPS provides linear power control, reduced circulating energy, and lower conduction losses, making it well suited for high-current regulation in the CC stage. As the output power decreases during the CV stage, DPS offers a wider soft-switching margin, reduced reactive power, and lower switching losses compared to SPS or EPS.

By applying EPS and DPS within their respective optimal operating regions, the proposed hybrid strategy effectively exploits the complementary advantages of both modulation schemes. Consequently, charging performance is improved over the entire charging range, with reduced total loss and stable CC–CV regulation.

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